

PTI

2021

CHAMOLI *Uttarakhand*

CAUSE: A large piece of rock and ice broke off a hanging glacier near Nanda Devi, became an avalanche

AREAS AFFECTED: Rishiganga and Dhauliganga valleys, Tapovan–Reni sector, Chamoli district

IMPACT: More than 200 dead/missing; Rishiganga hydropower project destroyed; Tapovan–Vishnugad hydropower project severely damaged; bridges and roads washed out



2023

SOUTH LHONAK, *Sikkim*

CAUSE: GLOF at South Lhonak Lake's moraine dam

AREAS AFFECTED: Teesta River valley—from Mangan district (Chungthang) downstream through Sikkim into the Darjeeling hills, Bengal

IMPACT: Over 70 dead; 1,200 MW dam at Chungthang destroyed; 33 bridges, 2 govt buildings, 16 roads damaged; NH10 (Sikkim's lifeline) cut off; 80% of Chungthang town severely damaged

2025 **DHARALI,** *Uttarakhand*

CAUSE: Sudden collapse of an exposed ice patch (cryo-hydrological event), traced by ISRO satellite analysis

AREAS AFFECTED: Bhagirathi valley, Uttarkashi district

IMPACT: At least 69 people reported dead. Bridges and irrigation canals destroyed





GETTY IMAGES

"As we raise the temperature, that glue starts to thaw. As it thaws, there's no more cement, no more glue to bind the rock. That's where the movement happens."

THE DANGER ZONES

Climate change is not just melting glaciers but already altering the structural stability of the Himalayan mountains themselves. Anil Kulkarni, the veteran glaciologist formerly with ISRO's Space Applications Centre and now associated with the Indian Institute of Science, warns: "With satellite imagery, ISRO has given the government a detailed summary of Himalayan glacial lakes, with movement and vulnerability. The downstream risk assessment has also been done in great detail, which ought to be used as the template for land-use planning. But are these reports studied when hydropower and urban infrastructure projects are planned?"

The Himalayan mountains supply not just water for drinking and farming, but also for energy. There are hundreds of hydropower projects in the Himalayan states, with an installed capacity of over 45,000 MW. India has consequently placed some of its most expensive infrastructure precisely where a glacial or debris flood will travel: on the valley floor. Hydropower projects need dams, tunnels, powerhouses, substations, worker colonies and access roads along rivers. Highways and pilgrimage facilities hug the same limited shelf of buildable land. One failure can add concrete, machinery and excavated muck to the surge; a blocked river or damaged reservoir can generate another pulse downstream.

The danger is clear and present. Sikkim is India's clearest warning. In October 2023, a rock avalanche struck South